

Ilja Safronovs

R&D Engineer — 3D reconstruction & computer vision · Giraffe360 · Riga, Latvia

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I work on 3D reconstruction and computer vision: Gaussian splatting pipelines, benchmarking systems that measure every change, and continuous research that turns the best new papers into production code. Financial Engineering student at RTU.

EXPERIENCE

Giraffe360 — R&D Engineer — Riga · 2025 – present

Solved blurry windows and mirrors in sparse-panorama 3DGS. instagram.com/p/DZsNRB8o772
A depth regularizer flattened outdoor views onto the wall. Fixed after 700+ experiments; in production since June 2026.

Shipped the looped drone video, from idea to production. instagram.com/giraffe360/reel/Dd9ODBVIHhE
I proposed it, agreed the look with designers, built the camera-path algorithm and shipped it; launched by Giraffe360, Oct 2026.

Led the team that built outdoor splats and orbital videos.
2 weeks to show first results, 3 weeks to push good ones to production. I led the team.

Influence accumulation: my densification method, still the team's best.
I found the research, implemented it, tested it and pushed it to production in Jan 2026. It is still the team's best densification method.

Implemented Analytic-Splatting in gsplat at CUDA level.
June 2026: exact per-pixel Gaussian integration, which reduces aliasing.

Built tools to measure 3DGS quality against known ground truth.
Validation tools and ground-truth datasets (July 2026) to check quality against known truth.

Mentored a junior engineer; together we replaced our depth algorithm.
Our depth estimates were poor. I mentored one junior engineer, and together we built a new depth algorithm (mostly his work): minima matching plus OMNI-DC to densify LiDAR depth.

Led the pipeline restructuring and created a 3DGS monorepo.
Led the restructuring (July 2026) and created a monorepo for all 3DGS extensions and client apps: ~3000-line files split, duplication removed, defects fixed.

Built an embeddable Rust/wgpu splat viewer and a comparison tool.
Rust/wgpu viewer (May 2026) that any app can embed and control, plus a multi-view comparison tool (Jan 2026).

Prototyped mirror planes and a fly-in dollhouse video intro.
Two working prototypes (June 2026): mirror planes that render correct reflections, and a fly-in "dollhouse" video intro.

Worked on mesh generation and point-count reduction for viewers.
Mesh generation from splats, combined with Gaussian splats; a point-count reduction stage (May 2026) for viewers.

Made diffusion-based artifact fixing runnable on shared GPUs.
Made a diffusion-based artifact-fixing tool (Difix, then NVIDIA Fixer) run remotely on shared GPUs in Feb 2026.

Three hackathon ideas recognized; the looped drone shipped.
Internal hackathon, Feb 2026 (13 teams, 51 ideas): three of my ideas recognized, one top 20, two special mentions.

Built an MCP server for the team task board.
An MCP server (June 2026, Node.js) so AI agents and tools can read and write the team task board.

Earlier: a small LLM service that wrote tour narration.
Jul–Oct 2025: a small LLM service that wrote tour narration (OpenAI API, AWS, S3).
Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University — Professional Bachelor, Financial Engineering 2024 – 2028 (expected)

Maths, statistics and programming applied to finance.

Qualification: senior data analysis specialist. Maths, statistics and programming applied to finance.

Coursework: mathematical analysis and algebra, probability and statistics, econometrics, time-series analysis, stochastic analysis, Monte Carlo methods, portfolio analysis, actuarial mathematics, data structures and algorithms, databases/SQL, computer networks, programming (Java, C++).

AI Talent Program (Ponder) Nov 2024 – Jan 2025

Mākslīgā intelekta talantu programma

Selective program in C++, Python, maths and AI.

Selective program: C++, Python, maths and AI development with industry mentors. Theory of AI: its history, neural network architectures (CNNs, transformers, GANs, VAEs), computer vision and reinforcement learning. Led to the Giraffe360 role.

Riga State Classical Gymnasium — Programming and Mathematics track

Rīgas Valsts klasiskā ģimnāzija

State mathematics olympiads and the Small Mathematics University.

Specialised track in programming and mathematics (today named "Programming + AI"). Latvian state mathematics olympiads; completed the Small Mathematics University (Mazā matemātikas universitāte) at the University of Latvia.

PROJECTS

load-balancing-experiments — seven experiments on DNS, queueing and load balancing.

flow — offline workout tracker in Dart/Flutter, in daily use.

monte-carlo-finance — option pricing, volatility and risk models in Python.

Portfolio journey — this tap-driven WebGL site in three.js.

Offline Graphing Calculator — 4th place of 27 in a school competition.

crypto-quant-lab — 59 backtested trading-rule versions on 6 coins.

projects_with_NEAT — neuroevolution agents that learn Flappy Bird and Pong.

RTU_program_parser — finds extra RTU lectures that fit your schedule.

Snake + maze generator — maze generation, terrain and path-finding in Pygame.

bezier-curve-js — interactive Bezier-curve drawing tool.

sort_visualization — 16 sorting algorithms animated in C++/SFML.

SKILLS

3D & vision	Gaussian splatting, 3D reconstruction, computer vision, depth estimation, anti-aliasing, mesh generation, diffusion-based artifact fixing, PyTorch, CUDA kernels
Research	Paper-to-production, benchmarking and evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
Tools	Linux, Docker, Git/GitLab CI, ClearML, AWS S3, Pygame, Selenium, pandas
Other	OpenAI API / LLM prompting, neuroevolution (NEAT), three.js/WebGL
Spoken	Latvian, English, Russian